

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
2	(a)	(i)	$\text{Vol} = \frac{4}{3}\pi \times (0.938)^3$ (1) substitution either in mm or cm or m $\text{Vol} = \underline{3.457 \text{ or } 3.46 \text{ or } 3.5} \text{ [cm}^3\text{]} (1)$ $\% \text{ unc in } V = 3 \times 0.02 \times \frac{100}{18.76} = 0.32 \text{ [\%]}$ seen or $\% \text{ unc in } r = 0.1 \text{ \%}$ so if you cube r then $3 \times \text{uncertainty}$ so $0.32 \text{ [\%]}$ (1)	1	1 1		3	3	3
		(ii)	$\% \text{ unc in mass} = 0.5 \times \frac{100}{26.3}$ or shown = $1.9 \text{ [\%]}$ (1) $\text{Density} = \frac{26.3}{3.46}$ ecf = $7.61 \text{ [g cm}^{-3}\text{]} (1)$ Total $\% \text{ unc} = 2.2 \text{ [\%]}$ ecf on $\% \text{ unc in mass}$ (1) Density shown as $7.6 \pm 0.2 \text{ [g cm}^{-3}\text{]}$ ecf Accept 7.61 ± 0.17 (1) No more than 2 sig figs for the uncertainty.		4		4	4	4
	(b)		For the 1st mark: – $\% \text{ unc in vol} = 2.9 \text{ [\%]}$ or greater and $\% \text{ unc in mass}$ same as for Tomos / unchanged – $\% \text{ uncertainty in density} = 4.8 \text{ [\%]}$ calculated – [absolute] uncertainty for Tomos = $0.01 \text{ [cm}^3\text{]}$ and $\% \text{ unc in mass}$ is the same – calculate [absolute] uncertainty for Jerry = $0.37 \text{ or } 0.4 \text{ [g cm}^{-3}\text{]}$ 2nd mark: So [absolute] unc in density <u>greater</u> for Jerry or [absolute] unc in density <u>lower</u> for Tomos			2	2	1	2

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	(c)		Tomos – Only stainless steel [within uncertainty range] ecf (1) Jerry – [Uncertainty range includes] stainless steel [and tin] and iron (1) ecf on any visible calculation Answers must be consistent with candidate values			2	2		2
			Question 2 total	1	6	4	11	8	11